

A certificate-based audit of simple connectivity for an explicit model associated with Wuebben’s proposed exotic $S^2 \times S^2$ construction

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Abstract

We define an explicit marked surface-bundle surgery manifold V_{aud} modeled on the data in Wuebben’s proposed exotic $S^2 \times S^2$ construction. Four hash-identified finite presentations are proved trivial by replayable derivation certificates under a published certificate specification. Automated acceptance of the stored derivations additionally assumes that at least one checker conforms to that specification; the two shipped implementations were developed within the project and have not received an independent human line-by-line audit. Separate combinatorial and geometric arguments identify those presentations with $\pi_1(V_{\text{aud}})$, including the marked clutching, normal frontiers, commonly based peripheral curves, drilled transport relation, and product-to-Lagrangian framing bridge. Consequently V_{aud} is simply connected. Comparison with Wuebben’s fixed surgery member is isolated as a fourteen-clause Source Formalization D, with each clause tied to a target location, mechanical evidence, and its remaining non-mechanical content. The three downstream manifold conclusions are then conditional on that formalization and the cited classification, symplectic, and Floer-theoretic results. A reported contrary fundamental-group computation remains unreconciled; the paper lists all live failure locations rather than assigning the discrepancy in advance. Thus the source-independent audit theorem, not the conditional transfer to the proposed exotic manifolds, is the paper’s primary result.

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1 Introduction

This paper is primarily about an explicitly defined audit object. Its primary result is the source-independent theorem

$$\pi_1(V_{\text{aud}}) = 1.$$

The proof joins finite group certificates to a displayed geometric construction of V_{aud} . Comparison with Wuebben’s intended manifold is a separate conditional application, and the exotic-manifold conclusions are conditional context rather than coequal claims of this note.

The motivating application is the construction of [1], whose theorem-critical step is the fundamental-group computation. The manifolds there rest on a specified Lidman–Piccirillo

piece V , obtained by two Luttinger surgeries [9, 4] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [3]. Write σ for the free orientation-reversing involution of $\partial V = S_0^3(Q)$, Q the square knot, $Z = V \cup_\sigma V$ for the symplectic double, $W = V/\sigma$ for the quotient, and $Z'' = V \cup_{f \circ \sigma} V$ for the Lidman–Piccirillo regluing. Wuebben’s three theorems are: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to Kawauchi’s manifold B , and (B, W) is a homeomorphic, non-diffeomorphic pair distinguished by the sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$. All three rest on the simple connectivity of V . Wuebben fixes one geometric manifold: his $(m, n) = (0, 0)$ member, using two geometric $+1$ Luttinger surgeries and the y_1 -based Ax half-drift on T_α . The exponents ϵ_A, ϵ_B in his relation sheet are orientation and presentation conventions, not four geometric parametrizations; he allows either sign for each. We certify all four convention specializations, so in particular the relator sheet for the fixed V is included. The other three specializations are convention-robustness checks, not claims about three further manifolds. Wuebben’s own 4,096-system grid varies, besides these two exponents, the three meridian signs of his transport corrections, the left/right placement of those corrections, the choice of conjugating arc for the Bs correction, and the arc, sign, and placement of the T_β push-off. In the rebuilt model none of these is a free choice: the corrections, basings, and push-off are computed as literal certified loops (Section 5), so only ϵ_A, ϵ_B remain as conventions, and the four certificates cover his grid at the fixed member.

This note rebuilds the theorem-critical group calculation, audits the framing bridge that connects it to the symplectic surgery, and records the downstream proof dependencies. The implementation does not execute Wuebben’s scripts or import their intermediate data. A quarantined extractor reads the immutable target text, while the geometric model is constructed independently; their marked data are compared only after both have been frozen. Every mechanically comparable item agrees. Agreement of two formalizations, however, is not the same statement as identity with the target construction. Section 6 therefore replaces the former four omnibus assumptions by Source Formalization D, fourteen atomic clauses with exact target locations and explicit residual judgments.

For the notation below, “ $n = 0$ ” means Wuebben’s Ax half-drift label on T_α ; the adjacent “ $n = 1$ ” control replaces Ax by Ar^{-1} . The letters M, N denote based meridians of T_α, T_β , while A, B are the two base generators.

The paper has four evidence levels, displayed once in Table 1. The first two are contributions of this note; the third is a conditional comparison with the target version; the fourth records consequences originating in Wuebben’s work.

Statement	Status	Trust boundary
Finite Certificate Theorem	source-independent theorem about four byte-bound presentations	certificate specification; conformance of at least one checker for automated replay
Audit-Manifold Theorem	$\pi_1(V_{\text{aud}}) = 1$	displayed geometric arguments, finite geometric artifacts, and named standard topology inputs
Source-Comparison Proposition	conditional identification with Wuebben's fixed member	the fourteen clauses of Source Formalization D
Conditional target application	conditional recovery of Wuebben's three conclusions	Source Formalization D and the cited classification, symplectic, and Floer-theoretic results

Table 1: The four claim levels. No implication changes one evidence type into another.

2 The audit manifold and principal statements

Write $\Sigma_{1,1}$ for an oriented once-punctured torus with based oriented spine loops α, β .

Definition 2.1 (the audit manifold). Let F be a closed oriented genus-two surface with an ordered five-chain (a, b, c, d, e) whose regular-neighborhood complement is the disjoint union of two disks containing marked points p, O . Let φ_0 be the orientation-preserving involution reversing the chain, fixing p, O , and acting freely on c by a half-rotation. Let

$$\psi_0 = T_a \circ T_b,$$

with T_b applied first and both twists supported away from a collar of e . Form the marked bundle R_{aud} over $\Sigma_{1,1}$ by clutching the α - and β -mapping cylinders of φ_0 and ψ_0 to one common marked fiber.

Inside R_{aud} put

$$T_\alpha = c \rtimes_{\varphi_0} S_\alpha^1, \quad T_\beta = e \times S_\beta^1.$$

Fix the basepoint $q = (p, *)$. Let $A = [\bar{\alpha}]^{-1}$ and $B = [\bar{\beta}]^{-1}$ be the based inverse base loops. Base the oriented meridians M, N along the initial fiber arcs y_1 to c_y and s_2 to e_e , respectively. In the product normal boundaries let λ_α be the y_1 -whiskered push-off that lifts the inverse α direction and closes along the selected half of c , and let λ_β be the s_2 -whiskered push-off of the inverse β direction along e . Their based words Ax and $(r^{-1}Mr)B$ are conclusions of Lemmas 5.3 and 5.4, not part of this definition. Define V_{aud} by the product-framed fillings

$$M\lambda_\alpha = 1, \quad N\lambda_\beta = 1.$$

Reversing an allowed peripheral orientation changes the displayed exponent, not the underlying filling.

Theorem 2.2 (Audit-Manifold Theorem). The manifold V_{aud} is simply connected.

Theorem 2.2 is source-independent: it concerns the explicit object in Definition 2.1. Its proof combines the finite certificate theorem below with the geometric identification in Section 5. Only the later Source-Comparison Proposition transfers it to the object named in Wuebben's target version.

3 The finite certificate theorem

Theorem 3.1 (finite-certificate statement). Let S be the byte string

`verification/luttinger/sealed_transport/r_presentations.json,`

whose SHA-256 digest is

`1e5fc8cb6872a25fa45a227c1e1c207b288c861a3998fe3aae8722ea510c852c.`

Read a positive integer j in S as the generator g_j and a negative integer $-j$ as g_j^{-1} . The top-level fields `ngens=3` and `relators` define a 3-generator, 78-relator presentation Q . Take the four entries of the top-level array `paper_fillings` whose `half_drift` field is `"n0_y1"`; these are entries 0–3, and their `(sign_a, sign_b)` fields run over $(+1, +1)$, $(+1, -1)$, $(-1, +1)$, $(-1, -1)$ in that order. Append the two word lists of such an entry, of lengths 94 and 132, to the 78 relators of Q , and call the resulting 3-generator, 80-relator presentation $P_{\epsilon_A, \epsilon_B}$ for $\epsilon_A = \text{sign_a}$ and $\epsilon_B = \text{sign_b}$. Every one of these four explicitly defined presentations presents the trivial group.

Definition 3.2 (letters, words, and equation keys). Fix generators $g_1, \dots, g_n$. Source relators are signed words in $\{\pm 1, \dots, \pm n\}$, with j denoting g_j . Certificate records use the monoid alphabet $L_n = \{1, \dots, 2n\}$ and the decoding map

$$\text{dec}(2j - 1) = j, \quad \text{dec}(2j) = -j.$$

Thus the required inverse-letter table is $(2, 1, 4, 3, \dots, 2n, 2n - 1)$. If w is a signed word, write $\bar{w}$ for its reverse with all signs changed, $\text{fr}(w)$ for free reduction, and $\text{cr}(w)$ for the word obtained by additionally removing inverse first/last pairs until none remain. Define

$$\text{ck}(w) = \min_{\text{lex}} \{ \text{cyclic rotations of } \text{cr}(w) \text{ and } \overline{\text{cr}(w)} \},$$

with $\text{ck}(1) = 1$. For certificate words u, v put

$$E(u, v) = \text{ck}(\text{dec}(u)\overline{\text{dec}(v)}).$$

Equality of equation keys therefore means equality after free and cyclic reduction, cyclic conjugacy, and inversion. Each operation preserves the normal closure of the equation word.

Definition 3.3 (literal rewrite trace). Suppose records $R_0, \dots, R_{i-1}$ have been accepted, with $R_r = (\ell_r, \rho_r)$. A trace on a word w is a list of pairs (r, p) with $r < i$ and $0 \leq p \leq |w|$. At that step the contiguous subword beginning at p must equal ℓ_r literally, and it is replaced by ρ_r . No cyclic matching, inversion, or implicit free reduction occurs inside a trace. Write $T(w)$ for its final word.

Definition 3.4 (certificate grammar). A `luttinger-kbmag-proof-v1` certificate is a finite ordered list of records (ℓ_i, ρ_i) , together with source-binding metadata and identity roots. Every side is a word over L_n . The case index must be an integer in the range of the source filling array. The source digest, case index and slug, generator count, full relator list, and inverse table must agree exactly with the hash-bound input. A record has exactly one of four proof forms.

- (i) An *inverse axiom* has $\rho_i = 1$ and $\ell_i = a \iota(a)$ for the required inverse table ι .
- (ii) An *input-relator record* names an in-range source relator q_k and satisfies $E(\ell_i, \rho_i) = \text{ck}(q_k)$.
- (iii) An *overlap record* names parents $a, b < i$, an integer offset d with $-|\ell_b| < d < |\ell_a|$, and two traces. The two parent left sides must agree literally on the resulting nonempty overlap. Let w be the shortest word containing both occurrences. Replacing the occurrence of ℓ_a by ρ_a gives one branch w_a , and replacing ℓ_b by ρ_b gives the other w_b . If the recorded traces give $u = T_a(w_a)$ and $v = T_b(w_b)$, validity requires $E(\ell_i, \rho_i) = E(u, v)$.
- (iv) A *change record* names an old record $o < i$ and traces on its two sides. Their literal outputs must equal the separately stored `reduced_left` and `reduced_right` words, say u, v , and validity requires $E(\ell_i, \rho_i) = E(u, v)$.

Finally the root array has length exactly $2n$; its entry for each $a \in L_n$ is an in-range record whose two sides are literally $(a, 1)$. Full-inventory mode also requires one correctly slugged certificate for each of the eight source fillings, with no repeated index or slug and with the stated presentation dimensions.

Theorem 3.5 (certificate soundness). Let G be the group defined by the source relators for a selected filling. If a certificate satisfies Definitions 3.2–3.4, then G is trivial.

Proof. Induct over the records. An inverse axiom is an equality in the free group. For an input record, the equation word is a freely and cyclically reduced cyclic conjugate or inverse of a defining relator, so its two sides agree in G . Every trace replaces a subword by the other side of an earlier valid equation and therefore preserves the represented element of G .

For an overlap, both untraced branches come from one source word by applying one earlier equality, so they agree in G ; their traces preserve that equality. Equality of equation keys transfers the resulting equality to (ℓ_i, ρ_i) . For a change, the two sides of an earlier valid equation are separately rewritten by earlier equations, and the same equation-key argument applies. Thus every accepted record is an equality in G . The root records say literally that every generator letter and its inverse equal 1, so $G = 1$. $\square$

Reference verification algorithm. The normative specification is Definitions 3.2–3.4. For comparison with the implementations, the complete control flow is:

```

bind source bytes, SHA-256, case, relators, generator count, inverse
table;
for each record in order:
    validate both words over  $L_n$ ;
    dispatch to inverse-axiom, input-relator, overlap, or change;
    replay every trace using only earlier record numbers;
    compare the required literal outputs and equation keys;
check one literal [letter]  $\rightarrow$  [] root for every letter; accept.
```

The standalone normative text is `verification/luttinger/CERTIFICATE_SPEC.md`. The Python and Ruby checkers separately implement it using only their standard libraries. They share neither source code nor runtime, but both were produced within this project and have not received an independent human line-by-line audit.

Proof of Theorem 3.1. For each of the four selected entries, the certificate bundle prepared with this version contains a certificate bound to the displayed source digest, case index, slug, complete relator list, generator count, and inverse table. The Python and Ruby implementations both accept every record and all six identity roots. The four certificates retain respectively 6672, 2470, 3803, and 3620 records in the source order $(+, +), (+, -), (-, +), (-, -)$. Hence Theorem 3.5 applies to each selected filling. Separately, two standalone checkers replay the frozen 99,860 elementary Tietze eliminations that produced $\mathbf{S}$ from its serialized raw input. No geometric interpretation of either byte string enters this proof. $\square$

The finite theorem is now relative only to the published certificate specification and to the proposition that at least one implementation conforms to it. The rest of the paper addresses the mathematically different question of what the hash-bound presentations describe. The principal result there is Theorem 5.7, which identifies them with the explicitly defined V_{aud} . Proposition 6.3 is the separate comparison with Wuebben under Source Formalization D. Corollary 7.1 records the downstream consequences only after both levels have been stated. Table 2 keeps these evidentiary levels separate.

Claim	Public artifact	What replay establishes	Remaining human input
$P_\epsilon = 1$	derivation DAG	every retained equation follows from earlier equations; every generator equals 1	checker implementation conforms to Theorem 3.5
raw to reduced presentation	frozen Tietze trace	each recorded elimination is valid and all 89 tracked words are transported	the raw presentation describes the geometric complement
peripheral pairs	simplicial paths and normal blocks	paths, common whiskers, orientations, local flatness, and normal-circle fibers	the standard PL interpretations stated in the proofs
framing agreement	exact-rational identities and finite hypotheses	constructive collar normalization, chart calculations, and first-jet independence	the Lagrangian-neighborhood theorem and the interpretation of the named push-offs inside V_{aud}
$V_{\text{aud}} = \text{Wuebben's fixed member}$	source dictionary, independent extraction, mutation controls, and author-script comparison	every mechanically comparable convention agrees	Source Formalization D
three manifold conclusions	dependency ledger	hashes, finite calculations, hypotheses recorded, and acyclic dependency	Source Formalization D and the cited classification, symplectic, and Floer-theoretic results

Table 2: Evidence levels. A replayed finite derivation, a checked combinatorial construction, and a dependency audit are intentionally not treated as the same kind of evidence.

4 Geometric identification of the audit manifold

This section proves that the finite presentations of Theorem 3.1 are presentations of the explicitly defined V_{aud} . It makes no comparison with the target paper’s denotation; that logically separate step is Section 6.

The genus-2 fiber is realized as a regular neighborhood of the specified five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R_{aud} is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α , T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

Theorem 4.1 (geometric artifact theorem). For each row of Table 3, the checkers listed in the companion supplement accept the corresponding frozen artifacts. Acceptance proves the finite assertion in the second column under the displayed checker semantics. It does not, by itself, prove the topological or smooth conversion in the last column; those conversions are supplied in the proofs below.

Proof. Each checker parses its input, verifies the recorded source digest and dimensions, exhausts the finite incidences or identities specified in its source, and exits successfully only after the asserted predicates have all been tested. The public proof ledger binds the artifacts and checkers by digest and records an acyclic dependency from these finite facts to Theorem 5.7. Table 3 states the meaning assigned to each successful replay; Table S1 in the companion supplement gives the exact filenames, checker sizes, and audit status. The later lemmas supply the mathematical conversion from the finite meaning to the geometric conclusion. $\square$

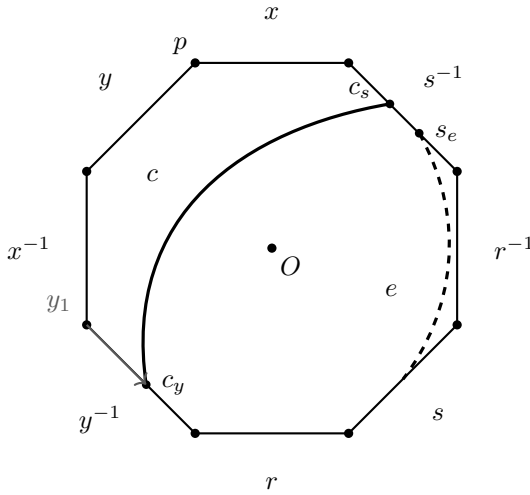
Table 3: Trusted-computing boundary for the geometric audit.

Layer	Successful replay establishes	Mathematical conversion still used
marked fiber	surface, five-chain, marked faces, rotation system, and involution satisfy the recorded finite tests	oriented ribbon classification and the periodic-disk theorem
relative clutching	based actions, seams, collars, and graph-bundle maps agree literally in two subdivisions	mapping-cylinder gluing and isotopy extension relative to collars
torus neighborhoods	every simplex-link pair, normal block, frontier comparability, and dual circle has the recorded finite type	PL local flatness and block-bundle interpretations
complement	the full-subcomplex retraction data, cellular presentation, Tietze moves, and 89 tracked words replay	cellular van Kampen
peripheral curves	common whiskers, normal circles, slopes, and both coordinate equalities are certified without filling relations	geometric interpretation as meridians and product push-offs

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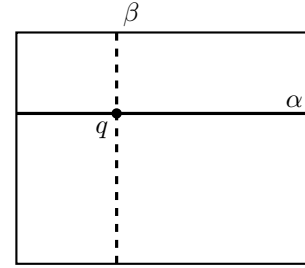
Table 3 continued

Layer	Successful replay establishes	Mathematical conversion still used
drilled transport	every simplex of the transport stack and homotopy grid misses both torus subcomplexes	simplicial homotopy in the complement
framing	relative normalizations, seam identities, symplectic-coordinate algebra, and first-jet isotopy identities hold	the Lagrangian-neighborhood theorem and compact push-off isotopy
PL-to-smooth bridge	the recorded relative collars, orientations, and compatibility hypotheses are present	standard smoothing and isotopy results for surfaces and surface bundles



(a) the marked fiber F

$$T_\alpha = c \rtimes_{\varphi_0} S_\alpha^1, \quad T_\beta = e \times S_\beta^1$$



base: once-punctured torus

(b) the two surgery tori

Figure 1: The marked data reconstructed by the audit. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent the basepoint p , while O is the center. Half-turn carries opposite edges to one another and induces $x \leftrightarrow r$, $y \leftrightarrow s$. The displayed arcs record the curve c , the five-chain curve e , and the initial y_1 whisker; the corresponding s_2 whisker is stored in the simplicial model. Panel (b) is a schematic of the two curve subbundles over the base loops. The geometric placements are checked as simplicial path and incidence assertions; the drawing is explanatory rather than an additional input to the computation.

The dictionary tested before Proposition 6.3 is displayed in Table 4; it is not reconstructed from the fact that the filled groups happen to collapse. Here “literal” means a stored edge path in the triangulation, with its basepoint and orientation retained.

Paper datum	Mathematical role	Certified realization
F, p, O	genus-two fiber with two fixed marks	86-vertex closed surface; both fixed vertices pass surface- and link-level tests
a, b, c, d, e	ordered filling five-chain	independently reconstructed ribbon; oriented c at V_2 represents $(rx)^{-1}$
φ_0	$x \leftrightarrow r, y \leftrightarrow s$	simplicial half-turn fixing p, O , checked on based paths
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}, y \mapsto yx, r \mapsto r, s \mapsto s$	ordered flip stack with T_b first; based monodromy replay
T_α	$c \rtimes_{\varphi_0} S_\alpha^1$	induced torus on the c -stack; local-flatness and subbundle checks
T_β	$e \times S_\beta^1$	induced product torus on the e -stack; local-flatness and product checks
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based edge paths from q , retaining their orientations
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers; all 592 normal fibers checked

Table 4: Marked-bundle dictionary for the fixed $n = 0$ member. Peripheral longitudes, which carry the sign-sensitive part of the identification, are displayed separately in Table 5.

Lemma 4.2 (equivariant marked fiber). There is an orientation-preserving homeomorphism $h : |F_K| \rightarrow F$ carrying the ordered curves (a, b, c, d, e) and marked points (p, O) to those of Definition 2.1 and satisfying $h\varphi_K = \varphi_0 h$. It carries the named based loops x, y, r, s with their displayed orientations.

Proof. Finite evidence. The independent ribbon reconstruction has

$$i(a, b) = i(b, c) = i(c, d) = i(d, e) = 1, \quad i(u, v) = 0 \quad \text{otherwise.}$$

Its regular neighborhood retracts to a graph with six vertices and ten edges, hence has Euler characteristic -4 . The two disjoint symplectic pairs (a, b) and (d, e) force genus two, so the neighborhood has two boundary components. The complement in the certified genus-two surface has Euler characteristic two and exactly those two boundary circles; it is therefore two disks. An exhaustive rotation-system check begins with the 36 cyclic orders at the independent four-valent crossings. Alternation, orientation preservation, two boundary faces, and preservation of the two faces leave exactly four systems; reversing the permitted curve orientations normalizes all four to one equivariant oriented ribbon type. The primary and independent fibers realize that same labeled type, including the involution on every directed curve end and the two p/O faces.

Topological input. Map vertex disks and edge bands by the checked cyclic correspondence. These product maps agree on attaching intervals, so they give an equivariant homeomorphism of the ribbon neighborhoods. Each complementary disk is invariant: if the involution exchanged them it would have no fixed point there, contrary to the certified two-point fixed set. The classical periodic-disk theorem identifies each nontrivial orientation-preserving order-two disk action with a half-turn. Extending over the quotient disks and lifting gives the required equivariant marked map. The based path comparison then reads, literally,

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y).$$

No fundamental-group filling or downstream theorem enters this argument. $\square$

Lemma 4.3 (relative monodromy and graph clutching). The map h can be chosen so that it conjugates the two monodromy representatives relative to the full c and e collars. It consequently extends to an orientation-preserving fiberwise homeomorphism

$$H : |K| \longrightarrow (R_{\text{aud}})_{\text{top}}$$

carrying the basepoint, both torus subbundles, and their product collars to the corresponding marked objects.

Proof. Finite evidence. The literal based actions are

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y), \quad \psi_0 : (x, y, r, s) \mapsto (y^{-1}, yx, r, s).$$

The second action is obtained from complete whiskered paths, not free homotopy classes, and is replayed in two differently subdivided beta stacks. On the three-row c collar the alpha map is exactly the free shift by four columns. The beta representative is the ordered supported word $T_a \circ T_b$, with T_b first; all 1,536 trace cells avoid the full e collar, and the 3,072 restricted tetrahedra are exactly its product staircase.

Topological conversion. Both bundle models are the union of a common fiber block and two mapping cylinders over the rank-two spine of $\Sigma_{1,1}$; there is no base two-cell. For a monodromy f , write

$$M_f = (F \times [0, 1]) / ((z, 1) \sim (f(z), 0)).$$

If $hf_K = f_*h$, the formula $[z, t] \mapsto [h(z), t]$ respects the seam and has the analogous inverse. For an isotopic target representative, the standard interpolating formula $[z, t] \mapsto [k_t(z), t]$, with $f_*k_1 = k_0f_K$, gives the same conclusion. The alpha conjugacy is exact on the c collar. For beta, both sides use the identical supported twist word; interpolating twist profiles is relative to the disjoint e collar. The two handle maps restrict to h on their common fiber and therefore glue, as do their inverses. This constructs H directly; bundle classification and Dehn–Nielsen–Baer are independent controls, not proof inputs. $\square$

Corollary 4.4 (the marked torus pair). Under H , the induced subcomplexes $c \rtimes_{\varphi_K} S_\alpha^1$ and $e \times S_\beta^1$ map to T_α, T_β , together with their product collars and the p -section. All smooth and symplectic arguments may therefore be performed in the audit-defined smooth target R_{aud} ; no smoothing of $|K|$ is chosen.

The model results through Corollary 4.4 concern V_{aud} and use no source-comparison clause. Five structurally different constructions test subdivision, monodromy order, twist sign, labels, and orientations. These internal controls support the audit object; they do not identify it with the object denoted by a source diagram. That comparison is made only in Section 6.

5 Peripheral data and the framing identification

Let $T = T_\alpha \sqcup T_\beta$ and let $C_{\text{aud}} = R_{\text{aud}} \setminus \text{int } \nu(T)$. This section identifies the normal boundary of each component, its commonly based peripheral pair, and the two filling slopes. The statements are deliberately separated so that a failure of one cannot be hidden by triviality of the final groups.

Lemma 5.1 (local flatness and the normal frontier). The two torus subcomplexes are disjoint, full, and locally flat. The barycentric level set

$$N_{1/2}(T) = \left\{ x : \sum_{v \in T} \lambda_v(x) \geq \frac{1}{2} \right\}$$

is an explicit normal D^2 -block bundle, and its frontier $F_{1/2}(T)$ is the normal-circle boundary. The derived frontier used by the computation is PL homeomorphic to $F_{1/2}(T)$, carrying every extracted dual loop to a literal oriented normal-circle fiber.

Proof. For every simplex $\sigma \subset T$, the complete link pair is checked to be

$$(S^3, \text{unknotted } S^1), \quad (S^2, S^0), \quad (S^1, \emptyset)$$

in dimensions 0, 1, 2, respectively. The enumeration covers all 296 vertices, 888 edges, and 592 triangles. Vertex-link 3-spheres and unknotted link circles carry independently replayed collapse and cyclic-knot witnesses. Thus the codimension-two local pairs are standard.

In a simplex with torus face A and opposite face B , barycentric coordinates give the unique block form

$$N_{1/2}(T) \cap \sigma \cong \Delta_A \times C(\Delta_B), \quad F_{1/2}(T) \cap \sigma \cong \Delta_A \times \Delta_B.$$

The formulas agree on common faces. If $b(s)$ is a mixed-simplex vertex of the computed derived frontier, the map

$$b(s) \mapsto \frac{1}{2}b(s \cap T) + \frac{1}{2}b(s \setminus T)$$

extends cellwise to the canonical subdivision map for $\Delta_A \times \Delta_B$ and hence gives the global PL homeomorphism. This is checked on 113,336 mixed vertices and 769,256 derived comparabilities. For every torus triangle the alternating dual loop has constant torus coordinate and is precisely the subdivided normal link circle. No regular-neighborhood uniqueness or smoothing theorem is used. $\square$

Lemma 5.2 (complement presentation). The induced subcomplex on the vertices outside T is a strong deformation retract of $|K| \setminus |T|$. Its maximal-tree edge generators and oriented triangle boundaries give a presentation of $\pi_1(C_{\text{aud}})$. The raw presentation has 99,863 generators and 321,702 relators; the sealed Tietze certificate reduces it to the 3-generator, 78-relator complement presentation in the source of Theorem 3.1, while transporting all 89 named peripheral and comparison words.

Proof. On each simplex outside the full subcomplex T , set the coordinates on T to zero and renormalize the remaining barycentric coordinates. The straight-line homotopy stays outside T , agrees on faces, and fixes the induced complement. A maximal tree on its 8,860 vertices has 8,859 edges; subtracting from the 108,722 complement edges gives the stated 99,863 generators. Collapsing the tree and applying cellular van Kampen to every oriented two-simplex gives the raw presentation. Each of the 99,860 recorded Tietze eliminations is replayed against the unique current relator that implies it; separate Python and Ruby checker implementations agree on the final presentation and every tracked word. $\square$

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction

package (the corrected relations of its §8.3 and the push-off basing correction of its §8.5) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper’s corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The closed section $\widehat{\Gamma}$ of the doubled bundle — the union of the two copies of the section through a fixed point of φ_0 — is extracted as a closed orientable genus-2 surface, and its self-intersection $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ is certified with an explicit normal push-off; this number enters the conditional consequences and the complete ledger in the supplement.

The relation sheet and the two fillings

To make Theorem 5.7 inspectable without opening the repository, fix the orientation sheet used by the triangulation and put $\delta = r^{-1}$. Before filling, the paper-coordinate presentation read from the literal paths has the surface relation, the following eight transport relations, and the drilled-fiber relation:

$$\begin{aligned} Ax A^{-1} &= r, & Ay A^{-1} &= s, & Ar A^{-1} &= x, & As A^{-1} &= Ny, \\ Bx B^{-1} &= y^{-1}, & By B^{-1} &= M^{-1}yx, & Br B^{-1} &= r, & Bs B^{-1} &= (r^{-1}M^{-1}r)s, \\ [x, y][r, s] &= 1, & B(s^{-1}r^{-1}yx)B^{-1} &= r^{-1}s^{-1}x. \end{aligned}$$

The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \quad \lambda_\beta = (r^{-1}Mr)B,$$

and the four convention sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \quad N\lambda_\beta^{\epsilon_B} = 1, \quad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \quad (1)$$

Changing an allowed orientation inverts the corresponding meridian or displayed longitude and changes the sheet label; it does not define a new geometric surgery. The finite presentations of Theorem 3.1 are the frozen Tietze images of these literal path presentations, but that geometric assertion is part of Theorem 5.7, not of the finite theorem.

Direction	Paper word	Stored based loop	Independent evidence
λ_α	Ax	1b_a_y1 , based by y_1 at c_y	literal path construction; 2,506-record complement certificate
λ_β	$(r^{-1}Mr)B$	1b_b_s2 , based by s_2 at s_e	punctured transport square and independent extraction; 1,540-record complement certificate

Table 5: The two sign-sensitive longitude identifications. Neither certificate contains a filling relation. Reversing an allowed orientation inverts the corresponding displayed element and is recorded by (ϵ_A, ϵ_B) .

Worked peripheral extraction

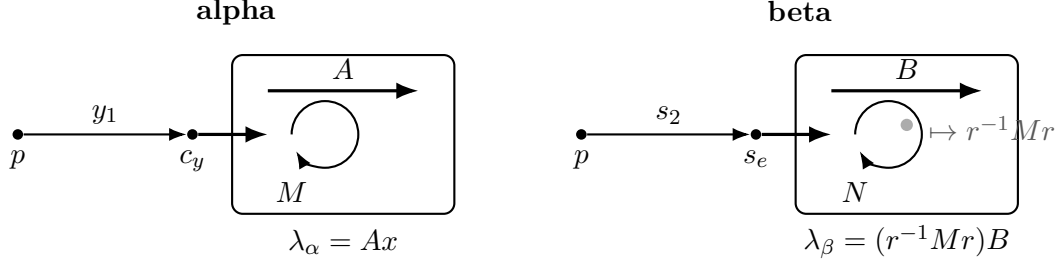


Figure 2: Common-whisker peripheral construction. On the alpha side the meridian and longitude use the same y_1 path to c_y . On the beta side they use the same s_2 path to s_e ; the swept basing square meets T_α once, and puncturing that intersection contributes the conjugated meridian $r^{-1}Mr$. The diagram records basing and word order, not metric geometry.

Lemma 5.3 (the alpha peripheral pair). Under the normal-frontier identification of Lemma 5.1, the stored loops `geom_M` and `lb_a_y1` are the commonly based meridian and product longitude (M, λ_α) , and

$$\text{lb_a_y1} = Ax \quad \text{in } \pi_1(C_{\text{aud}}).$$

The displayed equality is certified using only the 78 complement relators, with neither filling relation present.

Proof. Consider T_α . The curve c meets the oriented y edge once, at c_y . Starting at p , the stored path follows the initial segment y_1 to c_y , makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M . The longitude uses that identical whisker and normal jog, traverses the base direction A , and closes along the selected half of c . The stored words are `geom_M` and `lb_a_y1`; they are what M and λ_α denote throughout, and they are what Theorem 3.1 literally fills with.

That the two loops use the same whisker is not an inference but a fact one can read off the sealed transport input: there `geom_M` = $y_1 \mu_\alpha y_1^{-1}$ and `lb_a_y1` = $y_1 \lambda_\alpha^\circ y_1^{-1}$ with the *identical* four-edge whisker $y_1 = e_{24046e90328e79763e72645}$, seven- and six-edge circles $\mu_\alpha, \lambda_\alpha^\circ$, and fifteen and fourteen letters in all. So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the stored word `lb_a_y1` with the paper's symbol Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y , whiskered by y_1 , jogged to the normal-block frontier, lifting the base generator A , and closed along the selected half of c . Algebraically, the elementary elimination pass over the sealed presentation leaves the 72-letter residual `lb_a_y1`⁻¹ Ax , but a proof-producing completion reduces that exact word to the identity in 61 steps. Independent Python and Ruby verifiers replay the complete 2,506-record ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/alpha_residual/`. The supplement records the superseded development history; only the sealed certificate is used here. $\square$

Lemma 5.4 (the beta peripheral pair). Under the same normal-frontier identification, the stored loops `geom_N` and `lb_b_s2` are the commonly based meridian and product longitude (N, λ_β) , and

$$\text{lb_b_s2} = (r^{-1}Mr)B \quad \text{in } \pi_1(C_{\text{aud}}).$$

Again the equality is certified in the unfilled complement.

Proof. The beta-coordinate word $\mathbf{lb_b_s}2^{-1}(r^{-1}Mr)B$ likewise leaves a 113-letter residual under elementary elimination. The same proof-producing complement system reduces that exact word to the identity in 82 steps; separate Python and Ruby verifier implementations replay its 1,540-record ancestry cone. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/beta_residual/`. Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 3.1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e , the base loop B , and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the paper’s word for this direction is $(r^{-1}Mr)B$ and not B . The stored words are `geom_N` and `lb_b_s2`, exported without substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ is certified in the complement exactly as in the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words “meridian” and “push-off.” $\square$

Lemma 5.5 (the drilled-fiber relation). The based relation

$$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$$

holds in $\pi_1(C_{\text{aud}})$, not merely in the unpunctured mapping cylinder.

Proof. The transported loop κ_3 , of edge length 31, misses c but meets e once; sweeping it along the core of the beta curve would therefore meet T_β and would not prove the stated complement relation. The paper uses the specified parallel push-off. Along that push-off, the complete mapping-cylinder stack — 17,664 tetrahedra on 2,966 vertices — has no vertex on either surgery torus. A 2×35 simplicial homotopy grid at p joins the push-off basepoint loop to the core one and identifies it with the presentation’s whiskered generator; it too is disjoint from both tori. The based monodromy target then reduces to $r^{-1}s^{-1}x$.

The finite checker exhausts the stack, both torus vertex sets, every grid cell, and the target path. Thus the entire transport homotopy lies in C_{aud} . This is the construction-specific fact that was absent from the earlier mapping-cylinder-only calculation; no group simplification or low-index fingerprint is used to supply it. $\square$

Theorem 5.6 (product framing equals Lagrangian framing). For the normalized Thurston symplectic form on R_{aud} , the product-framed push-offs in Lemmas 5.3 and 5.4 represent the canonical Lagrangian-framing classes of T_α and T_β . In particular neither acquires a meridian component when transported into a Weinstein neighborhood.

Proof. The proof takes place in the already smooth target R_{aud} . By Corollary 4.4, the marked homeomorphism H transports the product collars and their push-offs from the PL model into that target. No smooth structure is placed on the source complex. A framing push-off below means the isotopy class, in $\nu(T_i) \setminus T_i$, of a sufficiently small copy of a curve on T_i .

Relative normal form. On an annular fiber collar write

$$\Omega = f(\theta, t) dt \wedge d\theta, \quad f > 0,$$

with core $t = 0$. Put

$$g(\theta, t) = \int_0^t (f(\theta, u) - 1) du, \quad H_s(\theta, t) = t + sg(\theta, t).$$

Then $\partial_t H_s = (1 - s) + sf > 0$. On one collar common to the compact core, H_s therefore has a smooth inverse in the radial coordinate. If $H_s(\theta, T_s(\theta, t_0)) = t_0$, then $T_0 = t_0$, $T_s(\theta, 0) = 0$, and

$$\partial_s T_s = -\frac{g(\theta, T_s)}{(1 - s) + sf(\theta, T_s)}.$$

Thus $(\theta, t_0) \mapsto (\theta, T_s)$ is the relative Moser isotopy, obtained by monotone inversion rather than an abstract flow theorem. At $s = 1$, differentiation in t_0 gives $f(\theta, T_1)\partial_{t_0}T_1 = 1$, hence $T_1^*\Omega = dt_0 \wedge d\theta$. The exact identities, fixed-core condition, and a uniform collar bound are replayed in `moser_cumulative_flow.py`.

For the alpha collar the deck involution is $\tau(\theta, t) = (\theta + \pi, t)$ and the quotient coordinate is $\bar{\theta} = 2\theta$; write π_α for the quotient map. The preceding radial map is defined directly upstairs by replacing $\bar{\theta}$ with 2θ in its coefficient. Periodicity makes it τ -equivariant, so no covering-lift choice is involved. Moreover

$$\pi_\alpha^*(dt \wedge d\bar{\theta}) = 2 dt \wedge d\theta = d(2t) \wedge d\theta,$$

which is the required factor-two normalization. These assertions are replayed in `equivariant_moser_lift.py`.

The two charts. Near T_β , ψ_0 is the identity near e , and the preceding normalization gives

$$\omega = dt \wedge d\theta_1 + K du \wedge d\theta_2, \quad K > 0;$$

the cotangent momenta are (t, Ku) . The in-fiber and base-normal push-offs are therefore constant-momentum copies. Near T_α , the mapping-torus seam is

$$(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u).$$

On its quotient put

$$\Theta_1 = \theta_1 - \frac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \frac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The Liouville primitive $t d\theta_1 + Ku d\theta_2$ forces these momenta by pairing with the Θ -frame; `weinstein_chart_check.py` replays that derivation, the seam, exterior algebra, and a second opposite-shear chart.

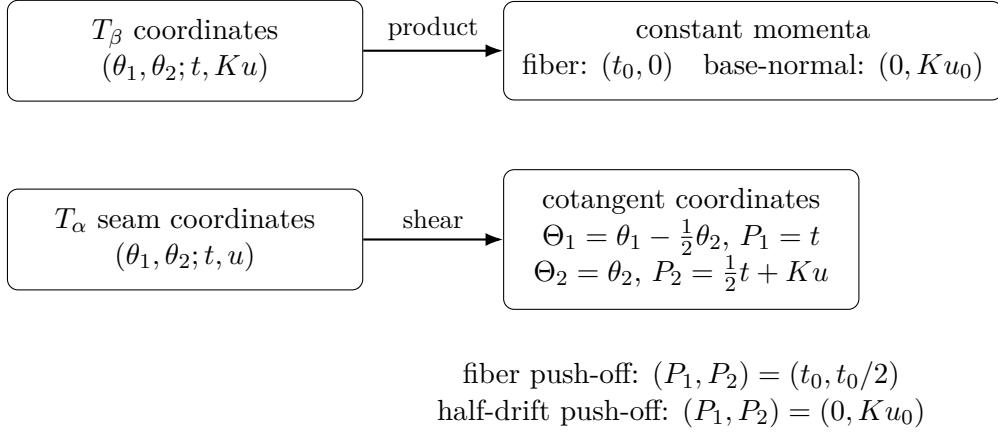


Figure 3: The two local Weinstein-coordinate calculations. Product push-offs become constant-covector copies in both charts, so the path from a push-off to the torus never winds in a normal circle. The chart-independence argument below shows that this boundary framing class is intrinsic.

Independence of the Weinstein chart. It remains to show that the constant-momentum conclusion is a framing class rather than a feature of the displayed coordinates. Given two Lagrangian-neighborhood charts, first compose with a cotangent lift so that their restrictions to the zero section agree. Their transition is then a symplectomorphism germ $\Phi : (T^*T, T) \rightarrow (T^*T, T)$ fixing the zero section pointwise. With fiber dilation $\delta_r(q, p) = (q, rp)$, define

$$\Phi_r = \delta_r^{-1} \Phi \delta_r \quad (0 < r \leq 1).$$

Although $\delta_r^* \omega = r\omega$, the two factors cancel, so every Φ_r is symplectic. Along the zero section the symplectic condition gives

$$d\Phi = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix}, \quad A = A^T.$$

Taylor expansion shows that Φ_r extends at $r = 0$ by the identity. Hence Φ_r is an isotopy of germs, relative to T , from the identity to the chart transition. Applied to a sufficiently small nonzero constant-covector copy of either named curve, this compact isotopy stays in $\nu(T) \setminus T$ and proves that the two push-offs have the same boundary class. The first-jet symplectic calculation is replayed in `weinstein_chart_independence.py`. The potentially dangerous momentum shift by a closed one-form could alter the pushed-off boundary class, but it moves the zero section unless the form vanishes and is therefore excluded here. The standard Lagrangian-neighborhood theorem, in the form used by [4, §2.1 and Proposition 2.2], supplies the existence of the charts; the argument above supplies the precise chart-independence consequence needed here. $\square$

Theorem 5.7 (audit-model identification). For each coherent convention sheet $(\epsilon_A, \epsilon_B) \in \{\pm 1\}^2$, there is a natural surjection

$$P_{\epsilon_A, \epsilon_B} \twoheadrightarrow \pi_1(V_{\text{aud}}).$$

In fact the map is an isomorphism. The four sheets express peripheral orientation conventions for the one audit-defined manifold V_{aud} , not four geometric targets. No clause of Source Formalization D enters this statement.

Proof. Lemmas 4.2 and 4.3 construct the marked bundle homeomorphism and carry the two curve subbundles and their product collars to the specified tori. Lemma 5.1 identifies their normal boundaries and meridians. Lemmas 5.3 and 5.4 identify the two commonly based product longitudes, including the conjugated meridian contributed by the punctured beta sweep. Lemma 5.5 places the final transport relation in the drilled complement. Theorem 5.6 identifies the product push-offs with the Lagrangian-framing push-offs.

Consequently attaching the two Luttinger solid tori adds exactly the normal closure

$$\langle\langle M\lambda_{\alpha}^{\epsilon_A}, N\lambda_{\beta}^{\epsilon_B} \rangle\rangle.$$

Cellular van Kampen therefore gives the asserted surjection from the filled presentation. Lemma 5.2 gives an actual presentation of the complement, not merely a list of valid relations; applying van Kampen with the two filling relations proves that the surjection is an isomorphism. Changing a common whisker simultaneously conjugates both elements of a peripheral pair, while changing an allowed orientation replaces the corresponding displayed exponent. Neither operation changes the filled manifold. This completes the identification with Definition 2.1 without comparing that definition to the data specified by Wuebben’s text and figures. $\square$

Proof of Theorem 2.2. Use any convention sheet in Theorem 5.7. Its source is trivial by Theorem 3.1; a quotient of the trivial group is trivial. Notice that this conclusion uses only the surjective part of Theorem 5.7 and no source-identification assumption. $\square$

6 Source Formalization D and conditional comparison

The finite certificate and Audit-Manifold Theorems are complete before this section begins. The remaining question is not “authorial intent” but a precise comparison of two mathematical specifications. We therefore freeze the following formalization of the target version.

Definition 6.1 (Source Formalization D). Source Formalization D is the conjunction of clauses D1–D14 in Table 6. Each clause gives an exact target location, the interpretation used to define V_{aud} , the available mechanical or cited support, and the assertion that remains non-mechanical. A clause marked “textual” is a transcription claim about the pinned target version; a clause marked “diagrammatic” or “smooth” is an additional mathematical identification.

Table 6: The source-to-audit dictionary. The last column, rather than the mechanical evidence column, is the conditional boundary.

	Target location	Audit interpretation	Mechanical or cited support	Residual assertion
D1	[1, Convention 2.1(C8), §7.1, Fig. 1]	the ordered curves are the five-chain (a, b, c, d, e) used in Definition 2.1	independent raw-text extraction; ribbon intersection and filling checks; source-figure hash	textual transcription of labels and order
D2	[1, Lemma 7.1]	the chain-reversing involution exchanges $a \leftrightarrow e$, $b \leftrightarrow d$, preserves c , and has marked fixed points p, O	exhaustive equivariant ribbon rotations; periodic-disk normal form; independent fiber subdivision	diagrammatic identification of the marked complementary disks

continued on next page

Table 6 continued

	Target location	Audit interpretation	Mechanical or cited support	Residual assertion
D3	[1, §7.1, Fig. 1] and the imported source figure	hidden, dashed, or paired segments continue the displayed curves rather than adding crossings	vector-source and planar-incidence extraction; mutation controls for added crossings	diagrammatic continuation convention
D4	[1]; Convention 2.1 (C2,C5); §§3.1, 7.2	$\psi_0 = T_a \circ T_b$ applies T_b first with the displayed right-twist normalization	based action computed in two subdivisions; reversed-order and reversed-sign controls	textual convention and its realization by the chosen supported representatives
D5	[1, Convention 2.1(C4), §7.3]	the cut square assigns φ_0 to the α holonomy and ψ_0 to the β holonomy	explicit graph-clutching map and inverse; seam checks	textual cut and orientation convention
D6	[1, §§7.2–7.3]	T_α is the c -over- α subbundle and T_β the e -over- β product subbundle, relative to the full c and e collars	collar-wise relative-monodromy certificates; disjoint induced-torus and local-flatness checks	identification of the target’s curve subbundles with these marked tori
D7	[1]; Convention 2.1 (C1,C3,C6)	paths compose left to right; $A = [\bar{\alpha}]^{-1}$, $B = [\bar{\beta}]^{-1}$; all peripheral loops use the stated common whiskers	literal path development and conjugation-sensitive calibrations	textual path and base-generator conventions
D8	[1, §8.3 and Table 1]	M, N are the oriented meridians based along y_1, s_2 , with the named correction paths	two independent peripheral extractors; local normal-circle fibers; punctured-sweep checks	identification of the target’s named basing arcs with the extracted arcs
D9	[1, §8.4]	the reference T_α base direction is the y_1 -based half-drift with word Ax	complement-only 72-letter identity certificate; side-sensitive mutation controls	textual choice of the reference half-drift
D10	[1, §9.1 and Table 1]	the target fixed member is $(m, n) = (0, 0)$ and uses the displayed F1 and F2 slopes	independent relation sheet extraction and exact comparison	textual member selection and relation-sheet transcription
D11	[1, Convention 2.1(C7), Remark 11.1]	ϵ_A, ϵ_B are relation-sheet orientation exponents, not four different geometric fillings	all four convention sheets certify the same filled audit object; sign-sensitive calibrations	textual status of the two signs and orientation matching
D12	[1, §8.7, Lemma 8.2]	the target uses the two smooth Lagrangian tori represented by the marked curve subbundles	local-flatness, normal-frontier, and topological-to-smooth transport certificates	smooth identification of the target tori with the audit tori
D13	[1, §§2.1, 8.7, 9]	the two geometric +1 surgeries use the Lagrangian-framing classes and orientations encoded by F1 and F2	two Weinstein-chart calculations; relative Moser checks; framing-class independence	correspondence between the target’s coefficient convention and the displayed attaching maps
D14	[1, §§7.1–8.7]	the marked bundle equivalence is represented by an orientation-preserving diffeomorphism relative to the c, e collars and boundary marking	explicit PL homeomorphism and inverse; relative collar data; standard smoothing of surface maps	smooth relative upgrade of the source-to-audit bundle equivalence

Several clauses are directly forced by the pinned text, while D2, D3, D6, D8, and D12–D14 retain diagrammatic or smooth comparison content. Splitting the former four omnibus assumptions into fourteen clauses prevents a verified textual choice from being bundled with a genuinely conditional smooth identification.

Lemma 6.2 (relative smooth source bridge). Under D1–D14, the marked bundle comparison is represented by an orientation-preserving diffeomorphism relative to the boundary marking and the two torus collars. It carries the two product-framed attaching maps of Definition 2.1 to the two Lagrangian-framed attaching maps of the target fixed member, and therefore extends across the two surgeries.

Proof. Clause D14 supplies the relative smooth representative of the marked bundle equivalence. Clauses D6, D8, and D12 carry the torus collars, common whiskers, and oriented meridians. Clause D13, together with Theorem 5.6, identifies the two longitude classes and surgery coefficients. Thus the boundary diffeomorphisms of each removed $T^2 \times D^2$ carry meridian and attaching slope to meridian and attaching slope. The product extension over $T^2 \times D^2$ on each filling piece gives the claimed diffeomorphism of filled manifolds. $\square$

6.1 The reported contrary computation and live failure locations

Wuebben’s public repository reports that Lidman and Piccirillo obtained a fundamental-group calculation disagreeing with his, currently summarized as $\pi_1(V) \neq 1$. Their published paper [3] proves $H_1(V) = 0$ and normal generation by $\pi_1(F)$ but does not publish this contrary presentation. No relation sheet, generator dictionary, or nontriviality witness for the reported calculation is available in the materials examined here.

The finite Theorem 3.1 is unaffected because its objects are explicit strings. If the contrary calculation concerns the same geometric member, the live explanations are deliberately left neutral: a clause of Source Formalization D may fail; a displayed geometric identification or geometric checker may be defective; the contrary calculation may be defective; or the two computations may concern different members or conventions. The absence of an external PL and symplectic audit prevents assigning the discrepancy to one location in advance. Relation-by-relation reconciliation requires the missing artifact.

Proposition 6.3 (Conditional source comparison). Under Source Formalization D, the marked smooth surgery data defining V_{aud} agree with the data defining Wuebben’s fixed $(m, n) = (0, 0)$ member. The resulting filled manifolds are orientation-preservingly diffeomorphic, with their boundary marking, peripheral curves, and framing classes identified.

Proof. Clauses D1–D3 and Lemma 4.2 identify the ordered marked fiber, the involution, and the complementary marked disks. Clauses D4–D6, Lemma 4.3, and Corollary 4.4 identify the two relative monodromies, the marked bundle, the two torus subbundles, and their collars. Clauses D7–D11 and Lemmas 5.1, 5.3, 5.4, and 5.5 identify the based meridians, product longitudes, drilled transport relation, convention sheets, and fixed member. Finally D12–D14 and Lemma 6.2 promote the relative comparison to the smooth category and carry the two attaching maps across the surgeries. The resulting diffeomorphism is the one asserted. $\square$

Corollary 6.4 (conditional simple connectivity). Under Source Formalization D, Wuebben’s fixed $(m, n) = (0, 0)$ member is simply connected.

Proof. Combine Theorem 2.2 with Proposition 6.3. $\square$

7 Conditional application to the target construction

This section records why the audit matters to the proposed construction; it is not part of the source-independent Audit-Manifold Theorem. Let V, σ, f, Z, W, Z'' denote the fixed member, involution, regluing map, and three manifolds in [1]. Source Formalization D is required even to identify V with V_{aud} , and the conclusions below also use named classification, symplectic, and Floer-theoretic results. Their exact statements, hypothesis checks, and the full dependency audit are collected in the companion supplement.

Corollary 7.1 (conditional application). Assume Source Formalization D and the classification, symplectic, and Floer-theoretic results cited in the companion supplement in the forms stated there. Then:

- (A) $Z = V \cup_{\sigma} V$ is homeomorphic but not diffeomorphic to $S^2 \times S^2$;
- (B) $W = V/\sigma$ is homeomorphic to Kawauchi’s manifold B , while the figure-eight knot is smoothly slice in B and not smoothly slice in W , so (B, W) is a homeomorphic non-diffeomorphic pair;
- (C) $Z'' = V \cup_{f \circ \sigma} V$ is homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$.

Proof. Under Source Formalization D, Proposition 6.3 and Corollary 6.4 supply the marked identification and simple connectivity required by all three arguments. For (A), the recorded intersection-form calculation and Freedman’s theorem give the homeomorphism type, while the symplectic and Kodaira-dimension inputs exclude the standard smooth structure. For (B), the Hambleton–Kreck inputs identify the quotient topologically. The figure-eight knot is smoothly slice in B by construction. The Klug–Levine and lifting alternatives obstruct a slice disk in W . It is not smoothly slice in W . For (C), the odd rank-two form gives the homeomorphism type and the cited Floer-theoretic mixed invariant distinguishes the smooth structure. The supplement states every external result used here and checks its hypotheses. Its proof ledger verifies the dependency bookkeeping; it is not an additional mathematical hypothesis. $\square$

8 Limitations and independent review

The project has three principal limitations. First, both theorem-critical filled-group checkers were produced within the same AI-assisted development process. Separate languages and implementations, negative controls, and the published mathematical specification reduce ordinary software risk but do not constitute an independent human line-by-line audit. Second, the marked fiber, peripheral, and framing arguments have been decomposed above so that a PL or symplectic topologist can audit them locally, but no such external audit is claimed. Third, the comparison with Wuebben’s fixed member remains conditional on every clause of Source Formalization D. The downstream corollary additionally depends on its cited external results. The reported contrary calculation in Section 6.1 is therefore an active reason to scrutinize both the source comparison and the displayed geometric arguments; it is not evidence against a particular one until its underlying presentation and dictionary are available.

The 100-case framing scan is not an answer to either limitation. All sampled shifted presentations were reported trivial, so the scan does not discriminate the correct framing and

has no logical role in Theorem 5.7 or Proposition 6.3. It is retained only as computational history in the supplement.

9 Conclusion

The revision isolates two theorem-level achievements. First, four hash-identified presentations are trivial under a complete published certificate specification, with the raw transport and all retained derivations replayable from frozen inputs. Second, those presentations identify the fundamental group of the explicitly defined V_{aud} , which is therefore simply connected. The marked fiber, relative clutching, torus collars, normal frontier, complement presentation, two peripheral pairs, drilled-fiber relation, and Lagrangian framing are separate results whose finite and conventional inputs are visible.

The remaining construction-specific uncertainty is Source Formalization D: whether the source text and diagrams define exactly the marked smooth data used for V_{aud} . The source comparison and the unavailable contrary computation meet at that boundary. The artifacts make its clauses finite and inspectable; they do not promote source interpretation into a certificate.

Even if a clause of Source Formalization D fails, the source-independent content does not disappear. The paper still supplies a complete certificate theorem for four explicit presentations, an explicit simply connected audit manifold, a proof-producing transport from a large cellular presentation, and a reusable separation between finite computation, geometric conversion, source comparison, and downstream citation. What would fail is the transfer of those results to Wuebben’s named manifold, not the results about the audit object itself.

10 Data and reproducibility

The target is Wuebben’s arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a.

The immutable Lidman–Piccirillo v1 source digest and all interpretation inputs are recorded in `TARGET.md`. The theorem artifacts, frozen inputs, certificate checkers, manifests, run transcripts, and reconstruction programs are in the public repository and its immutable v2.0.1 archival snapshot [2]:

<https://github.com/VentiMath/exotic-s2xs2-verification>,
<https://doi.org/10.5281/zenodo.22171309>.

The version DOI identifies the exact v2.0.1 verification snapshot; the concept DOI <https://doi.org/10.5281/zenodo.22169753> resolves to the newest archived version.

The submission release has three intentionally different replay levels. The certificate-only path reads frozen files and requires only standard Python and Ruby runtimes. The geometric-audit path recomputes the finite marked-fiber, torus, frontier, peripheral, and framing checks. The full reconstruction path additionally uses the locked GAP 4.11.1 and KBMAG 1.5.9 environment and the proof-producing KBMAG patch. Commands, expected record counts, negative controls, and hashes are listed in the companion supplement. The deterministic source and archive digests are recorded in `ARXIV_SUBMISSION.md`.

Three source-extraction replays additionally require locally supplied copies of third-party material that the release does not redistribute: Wuebben’s paper text, two files from his public code repository, and the Lidman–Piccirillo source figure. Their expected digests and exact retrieval instructions are in `verification/IMPORT.md`. The frozen extraction outputs, their comparison with the audit model, and every certificate used by the Audit-Manifold Theorem are contained in the release and replay without those third-party files.

The v2.0.1 release, PDFs, deterministic source archive, manifest, and clean-checkout replay were checked before that archival deposit was created. The present v2.1.0 revision changes the exposition and the formalized source comparison: it defines V_{aud} before the certificate machinery, replaces four omnibus assumptions by Source Formalization D, states the geometric trusted-computing boundary, makes the smooth relative bridge explicit, and rewrites the downstream result as a conventional conditional corollary. Before arXiv submission the v2.1.0 release must itself be archived and its exact version DOI added to the submission metadata. The finite filled-group certificates and the sealed transport are unchanged.

References

- [1] B. J. Wuebben, *An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$* , arXiv:2608.17267v1 (2026).
- [2] J. Clyde, *A certificate-based audit of the fundamental-group and framing computations in Wuebben’s proposed exotic $S^2 \times S^2$ construction*, verification artifact v2.0.1, Zenodo (2026), <https://doi.org/10.5281/zenodo.22171309>.
- [3] T. Lidman and L. Piccirillo, *Distinguishing closed 4-manifolds by slicing*, arXiv:2505.14387v1 (2025).
- [4] D. Auroux, S. K. Donaldson, and L. Katzarkov, *Luttinger surgery along Lagrangian tori and non-isotopy for singular symplectic plane curves*, Math. Ann. **326** (2003), 185–203.
- [5] A. Constantin and B. Kolev, *The theorem of Kerékjártó on periodic homeomorphisms of the disc and the sphere*, Enseign. Math. (2) **40** (1994), 193–204.
- [6] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 69, Springer-Verlag, 1972.
- [7] J. F. P. Hudson, *Piecewise Linear Topology*, W. A. Benjamin, 1969.
- [8] R. C. Lyndon and P. E. Schupp, *Combinatorial Group Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer-Verlag, 1977.
- [9] K. M. Luttinger, *Lagrangian tori in $\mathbb{R}^4$* , J. Differential Geom. **42** (1995), 220–228.
- [10] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), 357–453.
- [11] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group*, Math. Ann. **280** (1988), 85–104.
- [12] I. Hambleton and M. Kreck, *Cancellation, elliptic surfaces and the topology of certain four-manifolds*, J. Reine Angew. Math. **444** (1993), 79–100.
- [13] I. Hambleton and M. Kreck, *On the classification of topological 4-manifolds with finite fundamental group: corrigendum*, Math. Ann. **372** (2018), 527–530.
- [14] I. Hambleton, *Intersection forms, fundamental groups and 4-manifolds*, Proceedings of the 15th Gökova Geometry-Topology Conference (2008), 137–150.

- [15] C.-I. Ho and T.-J. Li, *Luttinger surgery and Kodaira dimension*, Asian J. Math. **16** (2012), 299–318.
- [16] T.-J. Li, *Symplectic 4-manifolds with Kodaira dimension zero*, J. Differential Geom. **74** (2006), 321–352.
- [17] W. P. Thurston, *Some simple examples of symplectic manifolds*, Proc. Amer. Math. Soc. **55** (1976), 467–468.
- [18] P. Ozsváth and Z. Szabó, *The symplectic Thom conjecture*, Ann. of Math. (2) **151** (2000), no. 1, 93–124.
- [19] P. Ozsváth and Z. Szabó, *Holomorphic triangle invariants and the topology of symplectic four-manifolds*, Duke Math. J. **121** (2004), 1–34.
- [20] P. Ozsváth and Z. Szabó, *Holomorphic triangles and invariants for smooth four-manifolds*, Adv. Math. **202** (2006), no. 2, 326–400.
- [21] A. I. Stipsicz and Z. Szabó, *On the minimal genus problem in four-manifolds*, arXiv:2307.04202 (2023).
- [22] M. R. Klug, *A relative version of Rochlin’s theorem*, arXiv:2011.12418 (2021).
- [23] J. Levine, *Polynomial invariants of knots of codimension two*, Ann. of Math. (2) **84** (1966), 537–554.
- [24] A. Kawauchi, *Rational-slice knots via strongly negative-amphicheiral knots*, Commun. Math. Res. **25** (2009), 177–192.
- [25] R. S. Palais, *Extending diffeomorphisms*, Proc. Amer. Math. Soc. **11** (1960), 274–277.
- [26] D. F. Holt, *KB MAG — Knuth–Bendix in Monoids and Automatic Groups*, software package, version 1.5.9.
- [27] The GAP Group, *GAP — Groups, Algorithms, and Programming*, <https://www.gap-system.org>.